

USER MANUAL

BARCC

Brain Atlas Regional Cell Counter

Version 8.08.000 - Generated July 29, 2026

BARCC is a specialized GUI application for analyzing immunofluorescence images. It enables researchers to overlay brain atlas sections, define regions of interest, perform automated cell counting, and export quantitative results.

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1. Introduction

BARCC (Brain Atlas Regional Cell Counter) is a specialized desktop application designed for researchers working with immunofluorescence (IF) microscopy images. It combines powerful image analysis tools with brain atlas registration to enable accurate, reproducible regional cell counting.

Purpose

The software allows users to overlay standardized atlas sections onto experimental TIFF images, define regions of interest through painting or atlas-based selection, apply sophisticated cell detection algorithms, and export quantitative results to Excel for further analysis.

Key Capabilities

- Import TIFF images and multi-page PDF atlas files; browse folders with progress tracking
- Allen Mouse Reference Atlas plates with semi-auto stitch (Reflect / move / rotate hemispheres)
- Interactive atlas alignment: Fit to Image, crop, move, rotate, scale; per-region edit tools
- Portable .catlas schematics: save labeled atlas + paint once, apply across channels
- Next Channel workflow keeps atlas and regions while switching fluorescence channels
- Freehand paint regions that register for counting (including on atlas coordinates)
- Configurable cell detection; manual add/remove/split; save/load cell masks across channels
- Random null cell distributions (optionally stratified by atlas region)
- Axons and Nets: regional intensity with background subtraction and counterstain normalization
- Perineuronal (PNN) shells (150% cell area) and intensity export (true + random)
- Automated regional cell counting with Excel export under an output/ folder

NOTE

BARCC is particularly valuable for neuroscience studies involving c-Fos, NeuN, axon/PNN markers, or other signals where regional quantification relative to brain anatomy is required. Multi-channel workflows (e.g. label on DAPI, quantify on other channels) are first-class.

What's New in Version 8.08.000

BARCC 8.08.000 is a major multi-channel atlas and analysis release: Allen atlas integration, portable .catlas files, cross-channel cell masks, axon/PNN intensity with corrections, random null distributions, and perineuronal shell measurement.

Allen Mouse Atlas & hemispheres

- Atlas > Import Allen Atlas: plate browser with Nissl reference strip and borders-only movable overlay.
- Semi-auto stitch editor: Reflect right-to-left, move/rotate halves, then Load into BARCC.
- Fit Atlas to Image, crop/move/scale/rotate with model-space placement (stable under zoom).
- After Reflect / bilateral stitch, structure names use _r and _l suffixes (e.g. V2M_r / V2M_l) so left and right are distinct in Atlas Manager and Count Cells.
- Download Full Allen Atlas option for offline/cache use.

.catlas schematic (multi-channel atlas)

- Atlas / File > Save Atlas Schematic and Load Atlas Schematic save a lossless .catlas package: zone mask, structure drawings, painted regions, Atlas Manager names, and placement (img_x/img_y).
- Label on counterstain (e.g. DAPI), save .catlas, load onto other channels of the same section.
- When background size changes between save and load, layers and offsets scale together (avoids drift).
- Legacy .atlas files still open; new saves use the .catlas extension.

Next Channel & Clear Atlas

- File / File Browser / Atlas Manager > Next Channel loads a new TIFF while keeping atlas drawings, zones, names, paint, and placement.
- Normal Import TIFF / Next Image fully clears atlas (no ghost double overlays).
- Clear Atlas removes overlay and labeled regions but keeps the TIFF.

Paint on atlas & menus

- Painted regions use atlas coordinates when an atlas is loaded; named paint merges into the zone mask for Count Cells without wiping Allen structures.
- Mask menu renamed Cell: detection tools plus Counting submenu (Count Cells, Show Zone Labels & Counts).
- Axons and Nets menu for intensity, counterstain normalization, and perineuronal tools.

Intensity, cell masks, random null, PNN

- Measure Region Intensities with optional Xth-percentile background subtraction and counterstain normalization (file from Counterstain Normalization Measurement).
- Exports include Pre_Correction and Post_Correction mean/median columns; Excel under output/.
- Save / Load Cell Mask (.barccmask + PNG) across channels; loaded mask locks Count Cells (no re-detect until Show Mask).
- Generate Random Cell Mask: same count as ground truth, random XY; stratified by atlas region when a .catlas/atlas is loaded (red = true, cyan = random).
- Draw Perineuronal Masks: shell from cell edge out to a disk of 150% cell area; also for random cells if present.
- Measure Perineuronal Intensity: by-structure mean/SEM and median/SEM (true and random); per-cell tables with area + PNN intensity.

See release-notes-v8.08.000.md in the repository root for the full changelog. Count Cells still counts only the ground-truth cell mask (not the random null mask).

What's New in Version 8.07.000

BARCC 8.07.000 improved Save Flattened Image so exports match what you see after painting and counting.

- Flattened export now composites TIFF base + yellow zone fills + black paint boundaries + red cell outline rings (when available).
- Default name {tiff}_flattened.tif under the image output folder when possible.
- Hardened overlay steps so partial failures still save a usable base image.

What's New in Version 8.06.000

BARCC 8.06.000 is a stability and usability release focused on Count Cells reliability and the Zone Labels & Counts viewer.

- Show Zone Labels & Counts now works: Cell > "Show Zone Labels & Counts" opens a dedicated table window listing every zone name and its cell count for the current file. Data comes from the latest Count Cells run, saved .xlsx/.csv results in the TIFF folder, or defined zone names (with dashes if not yet counted). A total row appears when numeric counts are available. The window refreshes after counting and when switching files in the left browser.
- Fixed hard crash at end of Count Cells on Windows: Saving the automatic `\_masked.tif` overlay used `compression='tiff\_deflate'`, which can segfault some Pillow/libtiff builds. The masked image is now saved as a standard uncompressed TIFF (same filename, stable on all platforms).
- Fixed Count Cells loading full-resolution TIFFs into memory when the canvas was not yet laid out (1x1 pixel at startup), which caused extreme slowdowns and apparent crashes on large microscopy frames. TIFFs now scale to fit the viewer window using screen/window dimensions as a fallback.
- Count Cells no longer calls the full Stop Paint path mid-count (which triggered `save_state`, `save_paint`, menu corruption, and extra redraws). A lightweight finalize step commits paint strokes to zones without UI side effects.
- Faster, safer per-zone counting: when a final cell mask is already computed, the redundant second watershed pass is skipped (connected components are used instead).
- Additional guards: try/except around the full count pipeline with user-visible error dialogs; 2D mask shape validation; progress dialog always closes in a finally block.

Together these changes make Count Cells complete reliably on typical Windows installations and give users an immediate tabular summary of regional results.

What's New in Version 8.05.000

BARCC 8.05.000 fixed critical issues when loading painted region bundles (.barccpaint) onto TIFFs and running Count Cells.

- Fixed "ufunc 'less'" error when loading .barccpaint bundles with named painted regions (JSON string zone IDs vs uint8 mask labels).
- Fixed silent Count Cells failures after bundle load: numpy edge cases in marker drawing, progress dialog timing vs results popups, and missing feedback when save paths were unset.
- Zone ID keys normalized to int on every load path; manual cell-edit masks cleared on new TIFF load.

What's New in Version 8.04.000

BARCC 8.04.000 focuses on major improvements to the Paint tool workflow, undo reliability for custom regions, and precision editing of painted region boundaries.

- Full repeated undo support for painted regions: Each individual paint stroke (mouse-down to mouse-up group) now creates its own undo checkpoint at the start of drawing. Naming a painted region, Stop Paint (auto-naming), Count Cells (force conversion), border/edge deformations, and Move Selected on painted zones are all fully undoable (up to 40 levels in the bounded history).
- Undo button: Prominent <- Undo button added to the Atlas Manager ribbon header (always visible when ribbon is shown). Also available via Edit > Undo (with Ctrl+Z accelerator shown). Keyboard Ctrl+Z continues to work. The ribbon list/header and visual elements (including paint strokes) update immediately after each undo step.
- Painted region border/edge expansion with deferred commit: When "Border drag resize enabled" is active and a painted region is selected, live dragging near the edge (or using the red local edge segment) updates the yellow/orange highlighted zone mask in real time for preview. The original black drawn boundary line stays in its previous position during the drag (providing a clear before/after view).
- Press **Enter** (or keypad Enter) after adjusting the border to commit: The current mask contour is extracted, the stored painted zone outline points are updated to the final deformed shape, ``_rebuild_paint_layer_from_data()`` is called to re-rasterize a clean black boundary line from the new points (with proper caps and joints), and a full ``show_page()`` redraws it. This "bakes" the expanded painted region and refits the visible black outline exactly to the new mask shape. The mask/zone data was already updated live; Enter finalizes the visual boundary.
- "Border drag resize enabled" checkbox now starts **unchecked** by default (in the Atlas Manager ribbon under the selected region tools). You must explicitly check it after selecting a region before edge or one-sided border drag tools will activate. This prevents accidental activation of the advanced editing mode.
- Paint mode indicator: When the Paint tool is active (entered via Paint > Start Paint or equivalent), a clear "[Paint] PAINT ON" label appears in bold red in the ribbon header. The main window title also updates to include " - [Paint] PAINT MODE". The indicator returns to "Paint: off" (gray) on Stop Paint, tool switches, or other exits from paint mode. The indicator is visible even if you hide the full ribbon (title bar fallback).
- Edge/border manipulation, "Move Selected Region", and related per-region tools now work fully for painted regions (in addition to atlas regions) using correct coordinate mapping (background/image space vs. atlas page space). Deformations update both the mask (for counting and yellow tint) and, on Enter commit for painted, the black visual boundary.
- Undo stack and state hygiene improvements: Painted region creation (draw + name), deformations, and finalization no longer cause unexpected batch undos or stale banner entries. The Atlas Manager "Labeled Regions" list and header now stay perfectly in sync with the actual zone data and visuals after paint actions and undos. Mask pruning on undo prevents the orphan auto-registration logic from re-adding

removed painted zones.

- All new paint editing and undo operations participate in the existing undo (save_state at the right points) and correctly refresh the ribbon, canvas, and paint_layer.

These changes make precise, iterative editing of custom painted regions (with live mask preview and explicit commit for the visible black boundary) reliable and user-friendly, while keeping the powerful Atlas Manager ribbon workflow intact.

What's New in Version 8.03.000

BARCC 8.03.000 delivers a major new workflow for atlas-based region editing via the Atlas Manager ribbon, plus global quick adjustments, improved mode visibility, and numerous robustness fixes for mixed image+atlas workflows.

- Atlas menu reorganization: "Import Atlas" (and global Crop/Move/Rotate/Scale plus per-region tools) now lives under a dedicated Atlas top-level menu (moved from File menu for better discoverability).
- New Atlas Manager ribbon (collapsible/expandable via header arrow; fully toggleable on/off from View > "Show Atlas Manager Ribbon" check item). The ribbon is the central hub for atlas region work:
 - Header always shows the currently selected region name/ID (or "No region selected").
 - Global tools now implemented as checkboxes (Crop, Move) so the active mode is immediately visible. Checking enables the corresponding click-drag behavior on the canvas (whole-atlas operations); unchecking returns to normal selection/naming.
 - "Move Selected Region" checkbox: when a region is selected (orange tint), click+drag inside it to translate *only* that zone's pixels in the mask. The underlying atlas artwork, other regions, and background image remain fixed. This is ideal for fine local alignment corrections without disturbing global registration.
 - "Border drag resize enabled" checkbox: when checked (and region selected), clicking near the perimeter of the orange/yellow region illuminates a local red edge segment. Re-click the red line to reposition the editable window; drag the red to locally push/pull only that portion of the boundary (linear falloff weights preserve connectivity and overall region integrity). Live shape update of the tinted region during drag via efficient rasterization + partial refresh. Release commits the edit.
 - Persistent red edge highlight survives panning, zooming, page changes, and show_page() calls (unless explicitly toggled off by a short click on an already-illuminated edge).
 - Quick Global Adjust section (new): Rot +5° / -5°, Scale +5% / -5% buttons that apply to the entire current atlas page (base artwork + all zone masks). Plus "Dialogs..." shortcut to the full global Rotate/Scale settings dialogs.
 - Selectable list of all labeled regions on the current atlas page. Clicking any entry selects it for editing (orange tint, ribbon header update, ready for quick adjust, edge drag, or move-selected). List auto-syncs after naming, transforms, page changes, etc.
 - Per-region quick adjust (unchanged from prior work but now clearly labeled "Selected Region Quick Adjust"): Rot +/-5°, Scale +/-5%, and "Dialogs..." for the currently selected region only (centroid-preserving transforms via the existing _apply_transform_to_region machinery).
 - Automatic mutual exclusion between global modes and edge features: enabling the border-drag checkbox deselects global Move (and Crop); enabling global Move or Crop automatically unchecks the border drag. Region "Move Selected" also cooperates with edit mode bindings. This prevents confusing overlapping behaviors and makes the active toolset obvious from the checkbox states.
- Major improvements to per-region atlas editing: individual regions on an imported atlas can now be

selected (canvas click on named yellow area, list, or Atlas menu Select Region tool), then independently rotated, scaled, translated (via the move-selected drag), or locally deformed via the red-edge drag. The rest of the atlas and background stay in place. Transforms update both the visual tint and the underlying label mask used for counting.

- Robustness fixes across the board:
 - Atlas crop now correctly converts canvas rect to model (native) coordinates using `_canvas_to_atlas`, clamps to image bounds, rebases `img_x/img_y` so the cropped content stays visually in place, prunes orphaned zone names, and fully clears stale selection/edge state.
 - Load-order fixes: loading a TIFF/image *after* an atlas no longer leaves stale `selected_zone_id` or `selected_edge_full_contour` that would hijack global Move drags (via the priority checks in `drag_start/drag_move`) or cause `is_near` tests to behave unexpectedly. All image-load paths (`import_tiff` and file-browser `_load_tiff_file`) now perform complete selection/edge/region-mode clears in addition to the zone/mask wipes.
 - Edge grab hit-testing made forgiving: clicks that land on boundary pixels (`zid==0` at exact integer sample) but are near the current selected region's border (or an already-illuminated red) are treated as edge-grab intent instead of falling through to a name prompt.
 - Global Move (`edit_mode`) and per-region features now correctly delegate in the drag handlers so that "Move Selected Region" or edge grab work even if the global Move binding is active (priority checks on mousedown; motion delegation).
 - Many additional state hygiene improvements (clears on page change, deselect, import atlas, crop end, etc.) so that ribbon list, orange tint, red edge, and move/edge flags stay consistent.
 - Ribbon checkboxes (global Crop/Move + per-region Move Selected + Border drag) plus the selected region header and list now give immediate, at-a-glance feedback about the current editing context and which tools are armed.
 - All new per-region and global quick adjust operations participate in undo (`save_state`) and correctly update the ribbon list/header after transforms.

These changes make fine-grained, region-by-region correction of atlas registration practical while preserving the ability to do global alignment. The visual distinction between "I'm moving the whole atlas" vs. "I'm only moving/deforming this one region" is now explicit via checkboxes and selection highlighting.

What's New in Version 8.02.002

BARCC 8.02.002 is a patch release that fixes a long-standing source of confusion between the counts the user sees visually on the annotated image/mask and the numbers reported in the spreadsheet.

- Fixed mismatch between per-zone cell counts shown on the mask/image (yellow zone labels like "RegionName\n(NN)" when "Show Zone Labels & Counts" is enabled, plus visible red cell markers in the annotated image) and the numbers in the exported spreadsheet (Cell Counts sheet) or CSV fallback.
- Root cause: The cell-to-zone assignment inside `count_cells_in_zones`` (used for the DataFrame that feeds both the spreadsheet and `last_df``) and the on-screen label positioning in `show_page`` unconditionally applied the atlas overlay offset (`-img_x` / `+ display_img_x`` etc.).
- Painted regions (and the main experimental TIFF background) live in a 0,0-aligned coordinate system matching the cell detection mask. Atlas/PDF zones may come from a differently sized/positioned overlay layer. Any non-zero `img_x`/img_y`` (from panning the atlas, zooming, dragging, alignment, etc.) would shift the lookup, causing cells to be assigned to the wrong zone (or no zone) for counting purposes.
- Visual markers were drawn at the correct positions (using raw centroids), so users would see e.g. 30 red dots inside a painted region but the spreadsheet and labels would report a different number.
- Implemented a robust size-based heuristic:
 - Compare zone mask size to the main `background_image`` / cell mask size.
 - If they match (within tolerance): zones are paint-on-background or standalone TIFF -> use direct cell coordinates (no offset) for accurate assignment and place labels over the background layer at canvas (0,0).
 - If sizes differ (atlas overlay): fall back to the previous `img_x`/img_y`` offset logic to align with the placed atlas layer.
- The same logic was applied to zone label text placement so the yellow (count) overlays appear over the correct regions even when an atlas overlay has a non-zero offset.
- After re-running Count Cells, the per-zone totals in the spreadsheet now exactly match the number of cells the user sees marked inside each region on the image and in the on-screen labels.
- No change to pure atlas workflows, detection logic, manual edits, or cases where no overlay offset is present. Version in exported settings JSON is "8.02.002".

This resolves the final major "numbers don't match what I see" issue for users relying on painted custom regions for quantitative analysis.

What's New in Version 8.02.001

BARCC 8.02.001 is a targeted patch release that resolves the last reported edge cases in the Paint tool for custom regions and improves dialog usability:

- First-paint reliability: Drawing a region, naming it immediately via right-click, and clicking Count Cells now succeeds on the **very first attempt** after loading any image (no more "No Regions Defined" for the initial named zone, with only later zones appearing in the spreadsheet).
- Fixed root cause in the 'img' bake path: The first ``stop_paint()`` (auto-called by Count Cells) + its ``show_page()`` would activate ``atlas_filetype='img'`` (via ``save_paint()``). ``load_page_image()`` then unconditionally reset ``zone_names[page]``, ``mask_images[page]``, and ``zone_counters[page]`` the first time ``page_images`` was populated for non-PDF content. This destroyed zones just registered by ``name_painted_region`` -> ``_convert_named_paints_to_zones``. A guard now restricts the destructive per-page init to ``atlas_filetype == 'pdf'`` only; 'img' (baked paint) and 'png' (loaded paint) cases preserve existing zone/mask state.
- Additional conversion hardening (building on 8.02.000 durability work): broadened stroke collection in ``_convert_named_paints_to_zones`` (always tries ``paint_group_data`` model points + canvas + last-resort any paint items for still-named groups), conditional ``dtag`` only after successful retirement in ``name_painted_region``, pre-clear re-try collection inside ``stop_paint``, and an ultimate force-add of lingering data groups before the error guard in ``count_cells``.
- Brightness slider dialog X button fix: The titlebar X (and Alt+F4) on the Brightness Settings window (containing the live brightness scale/slider) now properly closes the dialog. Previously wired to a no-op ``disable_event``. Applied the same ``window.protocol("WM_DELETE_WINDOW", window.destroy)`` pattern (already used by Mask Settings) to Brush Settings, Scale Settings, and Rotate Settings for consistency. Transparent window registration and `<Destroy>` cleanup continue to work. Progress dialogs remain hardened (defensive no-op flag) so early close cannot crash counting/detection.
- All prior v8.02.000 guarantees (immediate naming registers zones, Count auto-stops paint, durable model coordinates, ``binary_fill_holes`` + neighborhood lookup for accurate interiors, retirement to prevent dups, full wipe on every new image load, auto Save Paint Layer into the left File Browser directory, etc.) now apply reliably even to the first painted+named region on a fresh load.

These changes complete the Paint tool reliability story. The workflow "draw, right-click name immediately after the stroke, click Count Cells (without pressing Stop Paint)" is now fully production-ready on the first try.

What's New in Version 8.02.000

BARCC 8.02.000 delivers major stability, accuracy, and reliability improvements focused on the Paint tool workflow for creating custom analysis regions:

- Painted zones that are named immediately after drawing (via right-click) now correctly and reliably register for cell counting. Clicking Count Cells automatically stops paint mode and converts all strokes - both user-named regions and auto-defaulted "Painted Region N" entries.
- Complete prevention of cross-image contamination: every new TIFF loaded (via File Browser or Import) performs a full wipe of all mask_images, zone_names, zone_counters, named_paint_groups, and durable paint geometry data.
- Durable storage of paint stroke geometry using stable model/image coordinates (not view-dependent canvas coords). Hand-drawn regions now survive zooming, panning, and internal canvas refreshes.
- Accurate interior filling of painted structures using `scipy.ndimage.binary_fill_holes` after boundary stroking. Cells that are visibly inside your drawn outlines are now counted correctly instead of only thin boundary pixels.
- Neighborhood search (5x5) around each detected cell centroid when looking up zone membership. This provides tolerance for drawing precision and any remaining micro-imperfections in region fills.
- Critical stability fix: Closing the "Counting Cells" or "Detecting Cells" progress dialog with the X button (or Alt+F4) before the operation finishes no longer crashes the application. All progress UI methods are now fully defensive no-op after early dismissal.
- Elimination of duplicate zone entries (e.g., 6 zones appearing when only 3 were drawn) through proper retirement of processed paint groups after conversion.
- Overall hardening of the entire paint -> named group -> zone mask -> counting pipeline.
- Paint menu reorganization: "Save Paint" moved from File menu to Paint menu and renamed "Save Paint Layer". New "Load Paint" command added.
- Save Paint Layer now auto-saves directly into the folder currently open in the left File Browser (with automatic unique naming to avoid overwrites), matching the auto-export style of Count Cells. No save dialog appears when a working directory is selected.
- Load Paint and Import Paint now default their file dialogs to the current directory shown in the left File Browser for faster workflow.

These changes make the Paint tool (for custom regions outside of atlas PDFs) fully trustworthy and production-ready for quantitative cell counting work.

What's New in Version 8.01.000

BARCC 8.01 introduces major improvements to the cell detection system and parameter tuning workflow:

- New default detection engine based on Laplacian-of-Gaussian (blob_log) blob detection. This method is significantly more robust on typical immunofluorescence images than the previous watershed approach.
- Full set of Blob Detection parameters now exposed in Mask Settings with live preview.
- New "Smart Suggest (Pre-tuning smart settings)" button - a completely local analysis tool that examines your image and current detections and recommends better parameter values (pre-tunes smart settings). No data is ever transmitted.
- Ability to instantly switch between the new Blob method and the legacy Watershed method.
- Autotune buttons now intelligently adapt their adjustments depending on the active detection method.
- When using Add Cell or Remove Cell, the Brush Settings dialog now opens automatically for immediate size control.

These changes dramatically improve the experience of tuning detection parameters for difficult or variable images.

2. Installation & Requirements

System Requirements

- Windows 10 or later (primary supported platform)
- Python 3.8 or higher
- At least 8 GB RAM recommended for large images

Image Format and Compatibility Requirements

BARCC is designed primarily for quantitative analysis of immunofluorescence images. The following specifications describe the supported and recommended image characteristics:

Supported File Formats

BARCC supports the following image file formats with varying levels of compatibility:

- TIFF (.tif, .tiff) - Fully supported and strongly recommended. Supports both single-page and multi-page (stacked) TIFF files. Handles 8-bit, 16-bit, and 32-bit floating point data. This is the only format recommended for serious quantitative work.
- PNG - Limited support. May be imported in certain workflows but is not recommended for primary analysis due to potential compression artifacts.
- JPEG / JPG - Not supported. Lossy compression makes these formats unsuitable for cell detection and quantitative analysis.

Bit Depth and Data Types

- 8-bit (uint8) - Supported
- 16-bit (uint16) - Fully supported and recommended for most immunofluorescence work
- 32-bit floating point - Supported (both 0-1 normalized and arbitrary ranges)
- Multi-channel images (RGB, RGBA, etc.) - Accepted but typically converted or reduced internally. Single-channel grayscale images are strongly preferred for best results.

Image Dimensions and Size

BARCC has no hard upper limits on image dimensions, but practical performance considerations apply:

- Recommended maximum: approximately 8,000-10,000 pixels on the longest side for comfortable interactive use.
- Very large images (e.g., > 12,000-15,000 pixels per side) may cause high memory usage, slow performance during detection, or instability during manual editing.

- File sizes above ~500 MB can lead to noticeable slowdowns in the Mask Settings dialog and when generating visualizations.

Resolution and Physical Scaling

BARCC is strictly pixel-based and does not read or use image resolution metadata (DPI or $\mu\text{m}/\text{pixel}$). All detection parameters (cell size, sigma values, peak intensity, etc.) are expressed in pixels. Users must convert biological size requirements into pixel values based on their specific imaging resolution.

Recommended Image Characteristics

For optimal performance with the current detection system (especially the Blob Detection method):

- File format: Uncompressed or lossless TIFF
- Bit depth: 16-bit or 32-bit floating point
- Background: Relatively dark and uniform
- Cell appearance: Locally brighter than background, roughly circular or elliptical
- Noise level: Moderate to low (use preprocessing denoising when necessary)
- Image size: 1,000-8,000 pixels on the longest side for best interactive experience

NOTE

While BARCC can technically load images outside these recommendations, results may be suboptimal. For best accuracy and performance, prepare images as single-channel TIFFs with good contrast between cells and background.

Installation Steps

1. Clone the repository:

```
git clone https://github.com/LaingLab/BARCC.git
cd BARCC
```

2. Install dependencies:

```
pip install -r requirements.txt
```

For full .xlsx export support (including the Detection Parameters metadata sheet) and the _masked.tif feature, also install the Excel engines:

```
pip install openpyxl xlswriter
```

For the best experience with automatic Excel exports (including a second sheet with all detection parameters), we also recommend installing the Excel engines:

```
pip install openpyxl xlswriter
```

If these packages are missing, BARCC will automatically fall back to saving results as a plain .csv file instead

of .xlsx. These packages are listed as recommended (but not strictly required) in the project's requirements.txt.

3. Launch the application:

```
cd Application  
python barcc.py
```

3. Getting Started

Upon launching BARCC you are presented with a large central canvas area and a menu bar across the top. The typical workflow follows these steps:

- Import a TIFF (File > Import TIFF or File Browser)
- Import an atlas (Atlas > PDF or Allen Atlas + stitch; optionally save .catlas)
- Align with Fit Atlas to Image, Move, Rotate, Scale, Crop
- Define regions via atlas click, Paint, or Load Atlas Schematic
- Optional multi-channel: Next Channel (keep atlas) or load .catlas / cell mask
- Configure detection (Cell > Show Mask Settings); edit/save cell masks as needed
- Count Cells and/or Axons and Nets intensity / PNN measurement
- Results auto-save under the image output/ folder

4. User Interface Overview

The main interface consists of a left file browser pane and a large central image canvas. The left pane lets you select a folder and browse all TIFF images within it. Double-clicking any file loads it as the active image. A checkmark column shows which images have already been counted (based on the presence of matching .csv or .xlsx result files).

All other functionality is accessed through the top menu bar. The interface is designed to keep as much screen space as possible available for the image and mask visualization.

Main Canvas

The large central area displays your experimental image with the atlas overlay on top (when loaded). You can pan using the scrollbars or Alt+drag. Use the mouse wheel to zoom in and out. Zoom is centered on the mouse cursor and keeps painted regions, atlas overlays, and masks perfectly aligned with the background image at all zoom levels.

Menus

File, Edit, Atlas, Paint, Cell, Axons and Nets, View, and related menus provide access to all features. The former Mask menu is now Cell (detection tools plus a Counting submenu). Many operations open auxiliary dialogs for parameter adjustment.

5. File Menu & Multi-Channel Workflow

Import TIFF

Loads the primary experimental image. Supported formats include single and multi-page TIFFs. After import, the image is automatically scaled to fit the window while preserving aspect ratio. Importing a new TIFF clears atlas and zone state (use Next Channel if you want to keep the atlas).

Next Channel (keep atlas)

File > Next Channel, the File Browser button "Next Channel (keep atlas)", or the Atlas Manager ribbon button opens a file dialog for another TIFF (e.g. the next fluorescence channel). Atlas drawings, zone masks, Atlas Manager names, painted regions, and placement offsets are preserved. Cell detection masks are cleared - reload a saved cell mask if you need the same detections. Autosave of counts/paint for the channel you leave is attempted when possible.

Save / Load Atlas Schematic (.catlas)

File or Atlas > Save Atlas Schematic writes a portable .catlas file under the image's output/ folder by default. Load Atlas Schematic applies that schematic onto the currently open TIFF without re-importing the Allen plate. Ideal workflow: label on DAPI, save .catlas, Next Channel to axon/PNN channels, Load Atlas Schematic if needed (or keep atlas via Next Channel).

Import Atlas (PDF) - under Atlas menu

Opens a PDF file containing brain atlas plates. BARCC uses PyMuPDF to render individual pages at high quality. You can navigate between pages of multi-plate PDFs. Allen Mouse Atlas import and stitch live under Atlas as well (see chapter 6).

Split Tiff

Utility for breaking apart stacked (multi-page) TIFF files into individual images. Useful when your source data contains multiple sections in one file.

File Browser (Left Pane)

BARCC v8.01 introduced a dedicated file manager pane on the left side of the main window. This makes it much easier to work with folders containing many TIFF images.

Click the "Select Folder" button at the top of the left pane to choose a directory. BARCC will scan the folder and display all .tif and .tiff files in a list. Double-click any file in the list to load it as the active working image.

A second column in the list displays a checkmark ([x]) next to any image that has already been processed. This checkmark appears automatically if a matching .csv or .xlsx results file (generated by Count Cells) exists in the same folder. This is very useful for tracking which images in a large dataset have already been counted.

The Refresh button rescans the current folder. This is useful if you add or remove files while BARCC is running.

NOTE

The File Browser works independently of the traditional "File > Import TIFF" menu. You can still use the menu for one-off files, but the left pane is much faster when working with a whole folder of images.

Save Flattened Image

Exports a composite of the TIFF, yellow zone fills, paint boundaries, and (when available) red cell outline rings as a TIFF (typically {name}_flattened.tif under output/). Useful for figure preparation and session records. Also used by autosave on image switch.

Previous / Next Image & Next Uncounted

Navigate the File Browser list (Ctrl+Left / Ctrl+Right / Ctrl+Shift+Right). These load a new section and clear atlas/zones unless you use Next Channel. Progress checkmarks reflect counted artifacts in output/.

6. Working with Atlas Sections

Accurate alignment of the atlas to your experimental image is critical for meaningful regional analysis. BARCC supports PDF atlas plates and Allen Mouse CCF plates, with tools centered on the Atlas Manager ribbon for global and per-region control.

Import Allen Atlas & stitch editor

Atlas > Import Allen Atlas opens a plate browser (coronal/sagittal). Preview Nissl + structure outlines, then either Load as drawn or Open stitch editor. The stitch editor Reflects the right-hemisphere Allen drawing to the left, then lets you move/rotate left, right, or both before Load into BARCC. Atlas > Download Full Allen Atlas pre-caches plates for offline use.

After bilateral Reflect/stitch, structure names include hemisphere tags: acronym_r and acronym_l (e.g. V2M_r / V2M_l). Left and right get separate zone IDs for selection and counting.

Fit Atlas to Image, Crop, Clear Atlas

- Fit Atlas to Image: resizes the atlas (and mask/borders) to the TIFF size and aligns top-left corners.
- Crop: draw a rectangle (aspect can lock to the TIFF); incomplete edge structures can be pruned.
- Clear Atlas: removes drawings and Atlas Manager regions while keeping the loaded TIFF.

The Atlas Manager Ribbon

The ribbon (View > "Show Atlas Manager Ribbon" to toggle visibility) is the primary interface for working with atlas regions. It is collapsible via the >/v arrow in the header.

- Header always displays the currently selected region (name and ID) or "No region selected".
- Global tools appear as checkboxes (Crop, Move) - their checked state shows at a glance whether whole-atlas click-drag modes are active.
- "Move Selected Region" checkbox: enables interior drag of the orange-tinted selected region to shift *only* its mask pixels (everything else stays fixed).
- "Border drag resize enabled" checkbox: arms edge-grab mode for the selected region.
- Global Quick Adjust: one-click Rot +/-5° and Scale +/-5% for the entire current atlas page (base + all masks), plus quick access to the full Rotate/Scale dialogs.
- Selectable list of every labeled region on the current page. Click any entry to select it for editing (orange highlight appears, ribbon header updates).
- Selected Region Quick Adjust: the same +/- rotate and scale buttons, but applied only to the currently selected region (centroid-preserving).
- All operations are undoable and immediately reflected in the ribbon list and on-screen tints.

NOTE

The ribbon automatically stays in sync after naming, transforms, page changes, crops, etc. Checkboxes enforce mutual exclusion (e.g., turning on border drag automatically unchecks global Move and vice versa) so you always know which tool family is armed.

Selecting Regions for Editing

Click a yellow (named) or orange (selected) area directly on the atlas canvas, or use the list in the ribbon, or Atlas menu > "Select Region" (temporarily rebinds clicks to pick a zone). The selected region receives an orange tint overlay and becomes the target for quick adjust, edge drag, and move-selected.

Canvas clicks on already-named regions now intelligently autoselect them into the ribbon (instead of re-prompting for a name). Clicks near the perimeter (even on boundary pixels) are treated as edge-grab intent when the border checkbox is on.

Global Quick Adjust (new in 8.03)

The ribbon now offers the same style of quick +/- buttons for the *entire* atlas page that were previously available only for individual selected regions:

- Rot +5° / Rot -5° - rotates the full rendered atlas artwork and all zone masks for the current page (expand=True semantics, size may grow).
- Scale +5% / Scale -5% - uniform scaling of the entire page content.
- "Dialogs..." shortcut to the full global Rotate and Scale settings dialogs (with numeric entry and axis-specific scaling).

These are the global equivalents of the per-region quick adjust. They affect every region on the page plus the underlying artwork; use them for coarse global corrections before fine per-region work.

Per-Region Editing & Quick Adjust

Once a region is selected (orange), you can:

- Use the Selected Region Quick Adjust buttons (Rot +/-5°, Scale +/-5%) - these transform only the chosen zone's mask while keeping its centroid roughly in place. The rest of the atlas and background are untouched.
- Open full per-region Rotate/Scale dialogs via the "Dialogs..." button (or Atlas menu).
- Enable "Move Selected Region" and drag inside the orange area to translate only that zone (see below).
- Enable "Border drag resize" for local boundary editing (see below).

Move Selected Region (translate only one zone)

With a region selected and the "Move Selected Region" checkbox checked, click and drag inside the orange area. Only the pixels belonging to that zone in the label mask are shifted; the atlas artwork itself, other zones, and the background image remain exactly where they are. This is perfect for nudging a single misaligned anatomical region without disturbing your global registration.

The checkbox and the global Move checkbox are coordinated so you can easily switch between moving the whole atlas layer versus moving just the current region.

Edge Grab & Local Deformation (expand/shrink from the border)

With "Border drag resize enabled" checked and a region selected:

- Click near the perimeter of the orange/yellow region. A local segment of the boundary is highlighted in bright red.
- The red segment is a movable "handle" centered on your click point (using a sliding window along the contour with falloff weighting).
- Click again on an already-red edge to re-center the editable window at that exact location.
- Drag the red line outward or inward. Only the vertices inside the local window move, with linear falloff so the deformation blends smoothly into the rest of the boundary. The rest of the region (and the rest of the atlas) stays fixed.
- You get live visual feedback: the tinted region shape updates in real time as you drag.
- Release the mouse to commit the new shape to the mask (used for counting) and the display.
- A short click (no drag) on a red edge toggles the highlight off.

The red highlight is persistent across pans, zooms, and show_page refreshes until you explicitly dismiss it. This gives you precise, one-sided control over region boundaries without having to repaint or globally scale/rotate.

Global Crop & Move (checkbox style)

The Global Crop and Move tools are now checkboxes in the ribbon (and check items in the Atlas menu). When checked, they rebind canvas clicks for whole-atlas operations (crop rectangle or drag-to-move the entire overlay layer via `img_x/img_y`). Their checked state gives clear feedback that you are in a global mode rather than a per-region editing mode. Unchecking (or enabling an edge feature) restores normal click behavior (naming / region selection / edge grab).

NOTE

Best practice for atlas work: Use global Crop/Move/Quick Rotate/Scale first for rough alignment of the whole plate, then switch to per-region tools (list or canvas selection + quick adjust + edge drag + move-selected) for fine anatomical corrections. The checkboxes and orange selection tint make the current context obvious.

Move Atlas (legacy / still available)

The classic click-and-drag of the atlas overlay (when the global Move checkbox is checked) repositions the entire atlas layer relative to the background. This remains the primary tool for coarse global alignment.

Rotate / Scale (global)

Use either the ribbon Global Quick Adjust buttons for small increments or the full dialogs (Atlas menu or ribbon "Dialogs..." under Global Quick Adjust). These affect the rendered atlas artwork and all zone masks on the current page. After a global rotate that expands the canvas, the layer offset is automatically handled on the next redraw.

Brightness & Contrast Adjustments

The atlas rendering can be lightened or darkened independently of the experimental image to improve visibility of boundaries during alignment.

NOTE

Best practice: Identify 3-4 reliable anatomical landmarks (e.g., ventricles, major fiber tracts, cortical boundaries) and align to those rather than trying to match the entire section at once. After global alignment, use the new per-region tools in the ribbon to tweak individual structures without disturbing the rest of the plate.

7. Paint Tools for Regions of Interest

When the standard atlas regions do not match your experimental needs, the Paint tools allow completely custom region definition.

Start / Stop Paint

Activates freehand drawing mode. Draw directly on the image with the mouse.

Pen vs Eraser

Switch between adding area (Pen) and removing area (Eraser). Brush size is adjustable via the Brushsize dialog.

Load / Save Paint Layer

Painted regions can be saved and reloaded in future sessions, enabling consistent analysis across multiple images or experiments.

The Paint menu now contains dedicated **Load Paint** and **Save Paint Layer** commands. When a folder is open in the left File Browser, Save Paint Layer automatically saves the current paint layer directly into that folder as `{image}_paint.png` (or with numbered suffixes to avoid overwriting). The left file list is refreshed automatically after saving. Both Load Paint and Import Paint default their file selection dialogs to the current working directory shown in the left File Browser.

Labeling Painted Regions

While in Paint mode, right-click any painted stroke to name it. BARCC treats each continuous drawing action (mouse down through release) as one structural boundary. Naming converts the group into a zone immediately for Atlas Manager and Count Cells. Stop Paint and Count Cells also auto-name remaining strokes as "Painted Region N".

- The connected group highlights in yellow when selected for naming.
- With an atlas loaded, paint uses atlas model coordinates and merges into the existing zone mask without overwriting other structure IDs.
- Named painted regions participate in cell counting and intensity export like atlas regions.

NOTE

Prefer closed loops so interiors fill correctly (flood fill + binary_fill_holes). After naming, the region appears in the Atlas Manager list.

8. Cell Detection, Masks & Editing

Cell detection tools live under the Cell menu (formerly Mask). Show Mask Settings opens the parameter dialog. Show Mask runs detection (or displays a loaded mask). Add / Remove / Split Cell and Finish Mask Edit support manual QC.

This is the most powerful and configurable part of BARCC. The Mask Settings dialog provides fine-grained control over cell detection. BARCC v8.01+ supports two different detection strategies that you can switch between at any time:

- Blob Detection (Recommended): Uses modern Laplacian-of-Gaussian (blob_log) blob detection. Generally provides the best results on immunofluorescence images with variably bright cells.
- Watershed (Legacy): The original threshold + distance transform + watershed pipeline. Retained for compatibility with older workflows.

You can switch between these two methods at any time using the radio buttons at the bottom of the Mask Settings dialog. Most users should use the Blob method for new work.

NOTE

A powerful new feature in v8.01 is the "Smart Suggest (Pre-tuning smart settings)" button. This fully local tool analyzes your current image and detection results and suggests better parameter values. No data ever leaves your computer.

Detection Method

At the bottom of the Mask Settings dialog you will find a clear choice between two detection engines:

- Blob (new/recommended): Modern multi-scale blob detection using Laplacian of Gaussian. Much more robust for typical fluorescent cell images.
- Watershed (legacy): The original method based on global/local thresholding followed by watershed segmentation.

When Blob is selected, the lower part of the dialog shows the Blob Detection parameters. When Watershed is selected, the legacy Watershed parameters are shown instead.

Blob Detection Parameters (Recommended)

These parameters control the modern blob-based detector:

Blob Threshold

The most important sensitivity control. Lower values detect more (and dimmer) cells but increase false positives. Higher values are more conservative. Typical useful range: 0.05 - 0.20.

Blob Min / Max Sigma

Controls the range of cell sizes the detector will look for. Min Sigma sets the smallest detectable feature size; Max Sigma sets the largest. For most immunofluorescence nuclei, Min Sigma around 1.5-3.0 and Max Sigma around 7-12 works well.

Blob Num Sigma

Number of scales tested between Min and Max Sigma. Higher values give finer granularity at the cost of speed. Default (12) is usually sufficient.

Blob Overlap

How much overlap is allowed between nearby blobs. Lower values reduce duplicate detections on clustered cells.

Blob Min / Max Area

Post-detection filters based on area (in pixels). Very effective at removing tiny noise blobs or huge clumps.

Blob Min Circularity

Requires detected blobs to be reasonably round. Raising this value helps reject irregular artifacts.

Threshold Methods

The Threshold Method controls how the image is converted to a binary (black and white) mask for cell detection. Four options are available:

- Otsu: Automatically determines the optimal global threshold using Otsu's method. Fast and effective on images with good contrast.
- Adaptive: Uses local areas to determine the threshold. Excellent for images with varying brightness across the field of view.
- Local: Similar to Adaptive but uses a different neighborhood computation. Useful when Adaptive produces too many or too few detections.
- Manual: Uses a fixed threshold value (0.0-1.0) that you specify. Provides maximum reproducibility across batches of images.

Cell Detection Parameters

These parameters control how individual cells are identified from the binary mask:

Manual Threshold

Range: 0.0 to 1.0. Only used when Threshold Method is "manual". Higher values make detection more

selective (fewer cells). Lower values increase sensitivity.

Adaptive Block Size

Must be an odd integer (e.g. 51, 101, 151). Size of the local region used for adaptive thresholding. Larger values consider more surrounding area. Smaller values are more sensitive to local changes. Recommended range: 51-151 pixels.

Local Radius

Integer value (typically 5-30). Size of the neighborhood for the Local threshold method. Larger values smooth noise but may miss smaller cells.

Min Cell Size / Max Cell Size

Integer values in pixels. Define the acceptable size range for an object to be counted as a cell. Min Cell Size filters noise and small artifacts. Max Cell Size excludes clumps. Typical values depend on magnification (e.g. Min 20, Max 100-200).

Circularity Threshold

Range: 0.0 to 1.0. How circular a region must be to be accepted as a cell (1.0 = perfect circle). Higher values enforce rounder shapes. Typical value: 0.7.

Min Peak Distance

Integer value (pixels). Minimum distance required between detected cell centers. Helps prevent over-counting of touching cells. Typical values: 5-10 pixels.

Peak Min Intensity

Range: 0.0 to 1.0. Minimum brightness required for a local maximum to be considered a cell center. Higher values detect only brighter cells. Lower values detect dimmer cells. Typical starting value: 0.1.

Watershed Compactness

Range: 0.0 to 1.0. Controls how the watershed algorithm separates touching cells. Higher values favor more compact, rounder boundaries. Lower values follow intensity gradients more closely. Typical value: 0.0.

Base Multiplier & Sensitivity Range

Advanced sensitivity controls. Base Multiplier sets overall detection sensitivity (default ~1.1). Sensitivity Range controls how much the sensitivity slider can influence results (default 0.2).

Preprocessing Pipeline

A full preprocessing chain can be applied before cell detection. Each step can be enabled or disabled independently. The dialog only shows parameters relevant to the selected methods.

Background Method

Options: tophat or none. Controls removal of large-scale background variations. Tophat uses morphological operations and is generally recommended. When enabled, the Ball Radius (structural element size, default 15) controls how large-scale the background variations removed are.

Denoise Method

Options: gaussian, median, bilateral, or none. Reduces noise before detection.

- Gaussian: Applies a Gaussian blur (controlled by Gaussian Sigma, typically 0.1-5.0).
- Median: Better at preserving edges (controlled by Median Kernel size).
- Bilateral: Preserves edges while smoothing (controlled by Bilateral Sigma Color and Bilateral Sigma Space).

Contrast Method

Options: stretch, clahe, gamma, or none. Enhances contrast before detection.

- Stretch: Simple linear contrast stretching.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): Local contrast enhancement. Controlled by CLAHE Kernel (typically 8-16) and CLAHE Clip Limit (typically 1.0-4.0).
- Gamma: Gamma correction. Values < 1 brighten the image; values > 1 darken it.

Enhance Method

Currently supports Unsharp Mask sharpening. Controlled by Unsharp Radius (typically 0.1-5.0) and Unsharp Amount (strength of sharpening, typically 0.1-5.0). Useful for making cell boundaries crisper.

Both manual mask editing overlays (red) and automatic cell detection masks remain visible and correctly aligned when you zoom in or out using the mouse wheel.

NOTE

Start with the default settings and adjust one parameter at a time while using "Show Mask" to preview the binary detection result. This iterative approach is the fastest way to obtain high-quality cell counts.

Autotune Panel

The Mask Settings dialog includes a convenient Autotune panel with one-click adjustments. These buttons intelligently adapt their behavior depending on whether you are using the Blob or Watershed detection method.

The available Autotune buttons are:

- More cells - Increases overall sensitivity. With Blob mode this primarily lowers the Blob Threshold and Min Sigma.
- Less cells - Decreases sensitivity and raises size/shape requirements.
- Bigger cells / Smaller cells - Adjust size-related parameters (Min/Max Area or Min/Max Cell Size).
- Brighter cells / Dimmer cells - Primarily adjust intensity sensitivity (Blob Threshold or Peak Min Intensity).

Note: The Autotune buttons are intentionally conservative. For best results on difficult images, use the new "Smart Suggest (Pre-tuning smart settings)" button instead (see below).

Export / Import Settings

You can now export your current detection and preprocessing settings as a portable .json file. This is useful for backing up configurations, sharing them with colleagues, or moving them between computers.

In the Mask Settings dialog, use the buttons at the bottom:

- "Export Settings..." - Saves your current Blob (or Watershed) configuration and preprocessing settings to a .json file of your choice.
- "Import Settings..." - Loads a previously exported .json file and applies all parameters immediately.

This is separate from the internal Presets system (which stores quick named presets locally in ~/.barc/presets.json).

Smart Suggest (Offline) - New in v8.01

This is one of the most powerful new features in BARCC 8.01. Clicking "Smart Suggest (Pre-tuning smart settings)" runs a fully local analysis on your current image and detection results. It then proposes specific parameter improvements with clear explanations for each suggestion.

- Everything runs 100% on your computer - no images or data are sent anywhere.
- Each suggestion has a checkbox. You can selectively choose which changes to apply.
- Buttons at the bottom allow you to "Apply All", "Apply All That Are Checked", or simply "Close".
- It works with both Blob and Watershed modes and gives context-aware advice based on your actual data.

This tool is especially useful when the simple Autotune buttons are too aggressive or not aggressive enough. It is the recommended way to get good starting parameters for new or difficult images.

Manual Cell Editing

No automated detector is perfect. Cell > Add Cell / Remove Cell / Split Cell / Finish Mask Edit let you correct

errors before counting. Brush Settings opens automatically for add/remove. Manual edits combine with the automatic mask: (auto | add) & ~remove.

Save / Load Cell Mask (cross-channel)

Cell > Save Cell Mask stores the current detection (including manual edits) as a portable .barccmask package (and a companion PNG) under output/. Cell > Load Cell Mask applies that mask to the current TIFF (resized if needed). While locked, Count Cells uses the loaded mask without re-running detection. Cell > Show Mask (with recalculation) unlocks and re-detects.

NOTE

Typical multi-channel flow: detect/edit cells on one channel, Save Cell Mask, Next Channel, Load Cell Mask, then Count Cells on the new channel with the same detections.

Random null cell mask

Cell > Generate Random Cell Mask creates a null distribution with the same number of cells as the ground-truth mask, at random XY locations (cell sizes shuffled). If an atlas/.catlas is loaded, counts are stratified by region: each structure keeps the same cell count as GT, placed randomly inside that structure. Display: red = ground truth, cyan = random. Count Cells does not count the random mask - only the true cell mask.

9. Axons and Nets (Intensity & PNN)

The Axons and Nets menu supports regional intensity measurement, counterstain normalization, and perineuronal (PNN) shell analysis on the loaded TIFF inside Atlas Manager regions.

Measure Region Intensities

Measures mean/median (and other stats) of image intensity inside each zone. A dialog asks whether to:

- Background subtraction: subtract the Xth-percentile intensity within each region (typical X = 5-20).
- Counterstain normalization: divide by Normalization_Factor from a file produced by Counterstain Normalization Measurement on the same atlas.

Exports under output/ as {name}_intensities.xlsx with Pre_Correction_Mean/Median and Post_Correction_Mean/Median (and detail columns). Corrected mean = $(\text{mean} - \text{BG}) / \text{factor}$ when both options are on.

Counterstain Normalization Measurement

Run this on the counterstain channel (e.g. DAPI) with the same .catlas/atlas. Saves {name}_counterstain_norm.xlsx where Normalization_Factor is the regional mean intensity (used later as a divisor for axon/PNN channels).

Perineuronal (PNN) multi-channel workflow

The recommended pipeline labels anatomy on the counterstain, detects cells on a cell channel, then measures perineuronal intensity on the quantification channel. Use the Axons and Nets and Cell menus as follows.

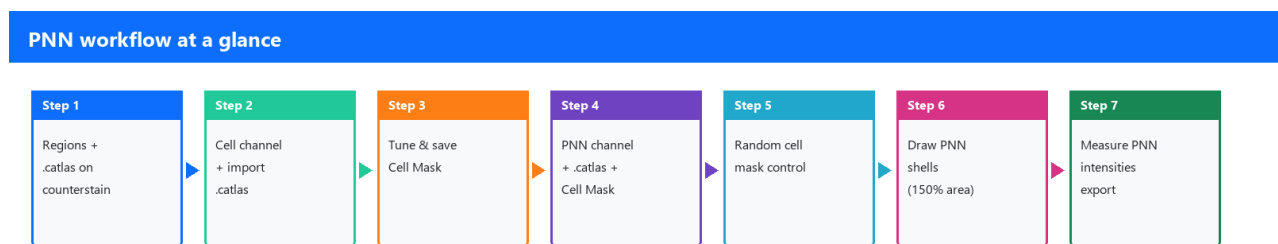


Figure. Seven-step PNN workflow at a glance (counterstain labeling through intensity export).

Step-by-step PNN procedure

Step 1 - Regions on counterstain; save .catlas

Open the counterstain image (e.g. DAPI / channel 0). Draw or import atlas regions (Allen plate, PDF atlas, or paint). Align with Fit / Move / Crop as needed. Save Atlas Schematic as a .catlas (cropped atlas) file under output/.

Step 2 - Cell channel + import .catlas

Open the image channel that contains the cells of interest (or use Next Channel while keeping the atlas). If the atlas is not already present, Load Atlas Schematic (.catlas) onto this channel so regions match the counterstain labeling.

Step 3 - Identify cells; save Cell Mask

On the cell channel, run Cell > Show Mask / Show Mask Settings. Tune detection parameters, and use Add/Remove/Split Cell for QC. When satisfied, Cell > Save Cell Mask (.barccmask / PNG) so the same detections can be reused on other channels.

Step 4 - PNN quantification channel; import .catlas + Cell Mask

Open the channel used for perineuronal-space intensity. Import the same .catlas (if needed) and Cell > Load Cell Mask so regions and cell locations match the prior steps. The loaded cell mask is locked for Count Cells / PNN until you re-detect with Show Mask.

Step 5 - Random cell masks (control distribution)

Cell > Generate Random Cell Mask creates a matched-pair null: each random cell has the same size/shape as one true cell, at a new XY. With a .catlas loaded, placement is stratified by region. Display: red = true cells, cyan = random. (Count Cells still uses only the true mask.)

Step 6 - Draw perineuronal masks

Axons and Nets > Draw Perineuronal Masks builds shells from each cell boundary out to a disk of 150% cell area (shell = outer disk minus cell bodies). If random cells exist, random perineuronal shells are drawn too. Display: magenta = true PNN; yellow = random PNN.

Step 7 - Measure perineuronal intensities

Axons and Nets > Measure Perineuronal Intensity measures mean intensity in each shell and writes spreadsheets under output/:

- {name}_pnn_by_structure.xlsx - one row per atlas structure: True/Random Mean, SEM_Mean, Median, SEM_Median
- {name}_pnn_cells_true.xlsx - one row per true cell: Cell_Area + Perineuronal_Intensity
- {name}_pnn_cells_random.xlsx - same for random cells (when random PNN masks exist)

BARCC Perineuronal (PNN) Analysis Workflow

Multi-channel pipeline from atlas labeling through intensity export

- 1 Draw or import regions**
 Align atlas with counterstain (e.g. DAPI).
 Save a .catlas (cropped atlas) schematic.
- 2 Open channel to quantify (cells)**
 Load the image channel containing cells of interest.
 Import the .catlas onto this channel.
- 3 Identify cells**
 Run Cell Mask / tune parameters on the cell channel.
 Save the Cell Mask for reuse.
- 4 Open PNN quantification channel**
 Load the channel used for perineuronal intensity.
 Import .catlas and the saved Cell Mask.
- 5 Random control cells**
 Generate Random Cell Mask (matched-pair sizes).
 Optional stratified null by atlas region.
- 6 Draw perineuronal masks**
 Shell from cell edge to 150% cell area.
 True (magenta) and random (yellow) shells.
- 7 Measure perineuronal intensities**
 Export by-structure mean/SEM & median/SEM.
 Plus per-cell true and random spreadsheets.

BARCC v8.08+ | Axons and Nets menu | See User Manual section 9

Figure. Detailed PNN workflow: (1) regions + .catlas on counterstain; (2) cell channel + .catlas; (3) tune and save Cell Mask; (4) PNN channel + .catlas + Cell Mask; (5) random control cells; (6) draw perineuronal shells; (7) measure and export intensities.

NOTE

Shell geometry: for cell area A , the outer disk has area $1.5 \times A$; the perineuronal mask is the ring between the cell body and that outer boundary. Re-draw PNN shells after regenerating random cells.

10. Counting Cells & Exporting Results

Once regions are defined and detection parameters are tuned, click "Count Cells" under the Cell menu.

To review results without re-running detection, use Cell > "Show Zone Labels & Counts". This opens a table window with Zone and Cell Count columns for the current TIFF. Toggle the same menu item again (or close the window) to hide it. Yellow on-image labels also appear when counts are in memory and the option is enabled.

BARCC will compute the number of detected cells within each named region. Results are now saved ****automatically**** (no file dialog) with the following files created in the same folder as your source TIFF:

- `YourImage.xlsx` - Excel workbook with two sheets:
 - Cell Counts - Region name, cell count, area, density, etc.
 - Detection Parameters - Complete record of every setting used (both cell detection and preprocessing). This is extremely useful for reproducibility and methods sections.
- `YourImage\_masked.tif` - The original image with the final cell mask (including any manual Add/Remove edits) drawn as a semi-transparent red overlay. Ready for figures or further analysis.

BARCC automatically saves two files when you click Count Cells (no manual Save dialog):

- `YourImage.xlsx` - Contains two sheets: "Cell Counts" (the actual results) and "Detection Parameters" (a complete record of every setting used for reproducibility).
- `YourImage\_masked.tif` - The original image with the final cell mask (after all manual Add/Remove edits) drawn as a semi-transparent red overlay. This is very useful for figure preparation.

To generate the .xlsx file (instead of falling back to .csv), the following packages are required:

```
pip install openpyxl xlswriter
```

These are listed as recommended (but not strictly required) in the project's requirements.txt. Without them, results are saved as a plain CSV file.

11. Saving & Export Options

- Flattened TIFF - image + zone fills + paint + cell rings (when available)
- Paint / .barccpaint - reusable paint + region bundles
- .catlas - full atlas schematic for multi-channel reuse
- .barccmask / cellmask PNG - portable cell detection masks
- Count Cells: {name}.xlsx (Cell Counts + Detection Parameters), _masked.tif, centroids CSV under output/
- Intensities: _intensities.xlsx, _counterstain_norm.xlsx
- PNN: _pnn_by_structure.xlsx, _pnn_cells_true.xlsx, _pnn_cells_random.xlsx
- Random cell mask: _random_cellmask.png (+ JSON meta)

12. Keyboard Shortcuts

Shortcut	Action
Ctrl + Z	Undo last action (paint, atlas edit, etc.)
Ctrl + S	Save flattened image
Ctrl + Left	Previous image in File Browser
Ctrl + Right	Next image in File Browser
Ctrl + Shift + Right	Next uncounted image
Enter	Commit painted border refit after edge drag
s (paint mode)	Toggle pen drag vs segment mode

13. Troubleshooting

Common Issues

- Cells not detected: Try lowering Peak Min Intensity or switching detection method / Smart Suggest.
- Too many false positives: Increase min size/sigma or use Remove Cell.
- Atlas drifts on zoom: Use current build (model-space img_x/img_y); re-Fit if needed.
- .catlas loads shifted: Prefer same-resolution channels; rebuild schematic after Fit; 8.08 scales layers with background size.
- Left/right structures not distinguished: Re-Reflect/stitch so names get _r/_l; too many structures may exceed uint8 bilateral split.
- Count Cells re-detects after Load Cell Mask: Ensure mask stayed locked; avoid Show Mask recalculate until you want a new mask.
- Random cells not in counts: By design - only the ground-truth cell mask is counted.
- Excel not written: pip install openpyxl; check the image folder output/ is writable.
- Count Cells crashes: Use v8.06+ (TIFF deflate fix); open TIFF after the window is laid out.
- Performance is slow: Close other apps; large frames use viewer downscale - 16 GB+ RAM helps.

Getting Help

For bugs or feature requests, please open an issue on the GitHub repository:
<https://github.com/LaingLab/BARCC>

Release notes for each version live in the repository root (e.g. release-notes-v8.08.000.md).

Thank you for using BARCC

We hope this software accelerates your research. Feedback and contributions are always welcome.